MILITARY SPECIFICATION

CANNON, 155 MM GUN M46

This specification is approved for use by all departments and agancies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of gun, cannon.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or requests for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-13931 - Cannons, Mortars, Recoilless Rifles and Recoilless Guns; General Specification for
MIL-P-14232 - Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage

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PUBLICATIONS

U.S. Army Armament Command

EPL 7238947 - Engineering Parts List, Cannon, 155MM Gun M46
DL 7238947 - List of Inspection Equipment for Cannon, 155MM Gun M46
Master LSQAP 7238947 - Master List of Supplementary Quality Assurance Provisions for Cannon, 155MM Gun M46
PDS 7238947 - Packaging Data Sheet, Cannon, 155MM Gun M46

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified in the contract, the contractor shall submit a first article. Unless otherwise specified (see 6.1.1), the first article shall be included in the first article package (see 5.1).

3.2 General requirements. Cannons and parts thereof shall conform to the drawings listed in Engineering Parts List EPL 7238947 and the requirements of this specification and MIL-C-13931.

3.3 Functioning. All parts shall function without interference, erratic movement or malfunction.

3.4 Headspace. Headspace, the distance between the front face of the closed firing block and the primer seat on the extractor shall be not less than 0.052 inch nor more than 0.065 inch.

3.5 Firing pin protrusion and retraction. When the firing mechanism is fired, the protrusion of the firing pin from the front face of the firing block shall be not less than 0.090 inch and not more than 0.107 inch. The firing pin shall be retracted behind the front face of the firing block when the firing mechanism is in the load position.

3.6 Performance.

3.6.1 High pressure resistance. The cannons shall be capable of withstanding chamber pressures of 54200 ± 1900 pounds per square inch when tested as specified in 4.5.5.

3.6.2 Material soundness. After completion of initial production or performance testing, all parts shall be free from cracks and fractures and shall comply with soundness requirements specified on the applicable drawings.

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3.6.3 Bore enlargement. Bore diameters, measured across the rifle lands after proof firing no more than 8 rounds, shall not exceed the basic (minimum) diameter specified on the applicable drawing by more than the amount specified in Table I.

Table I Bore enlargement

Distance forward of commencement of rifling	0.10 inch	1 inch	2 cal.	3 cal. to muzzle
Enlargement (inch)	0.006	0.006	0.003	0.002

NOTE: Interpolate to determine the allowable bore enlargement between the distances given.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used in this specification are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

4.4 First article inspection. A first article shall be submitted for inspection in accordance with contract requirements. The first article shall be representative of the production process to be used during quantity production. The first article shall be subjected to the inspections, examinations, and tests specified herein and such other inspections, examinations, and tests as is necessary to determine compliance with the requirements of the contract.

4.5 Quality conformance inspection.

4.5.1 Inspection provisions. Cannon shall be inspected for conformance to requirements of the provisions of this specification, MIL-C-13931, SQUAPs listed in Master LSQAP 11578365, and SQAP-APPENDIX-WVT.

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4.5.1.1 Final examination. Final examination of each cannon shall be completed after testing and prior to preservation and packaging to determine compliance with this specification, MIL-C-13931, and as specified in the contract. The inner and outer surfaces of each finish machined tube, the breechblock, bushing, breech ring and carrier shall be magnetic-particle tested in accordance with the applicable part drawing. The inner surface of the tube shall be examined by means of a borescope. Any evidence of cracks or fractures shall be cause for rejection.

4.5.1.2 Inspection approval stamp. The application of the inspection approval stamp shall be as specified in MIL-C-13931.

4.5.2 Inspection equipment. Data lists containing drawing numbers of Government designed inspection equipment used in the inspection of parts, subassemblies, and assemblies are listed in DL 7238947.

4.5.2.1 Acquisition, maintenance and disposition. Unless otherwise specified (see 6.1.1) responsibility for acquisition, maintenance, and disposition of acceptance inspection and test equipment prescribed on the List of Inspection Equipment DL 7238947, and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.2.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it shall be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.5.3 Examinations.

4.5.3.1 Headspace. The headspace shall be checked on each cannon with the firing mechanism mounted on the cannon. Gages 8778048 and 8778049 or approved equivalent shall be used to determine compliance with the headspace requirements.

4.5.3.2 Firing pin protrusion and retraction. The firing pin protrusion shall be measured on each firing mechanism before it is mounted on the obturator spindle of the cannon. Protrusion of the firing pin beyond the face of the block shall be measured with the firing mechanism in the closed and locked position. Indicator gage 8775230 or an approved equivalent shall be used to determine compliance with the firing pin protrusion requirement. The firing pin shall be visually checked for compliance with the firing pin retraction requirement, when the firing mechanism is in the load position. Failure of the firing mechanism to comply with the protrusion or the retraction requirement shall be cause for rejection.

4.5.3.3 Workmanship. Workmanship shall be examined visually in accordance with MIL-C-13931.

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4.5.4 Tests.

4.5.4.1 Functioning tests. Functioning tests shall be conducted in accordance with MIL-C-13931 and as specified herein. A minimum of two complete cycles of operation shall be performed on each cannon.

4.5.4.2 Firing mechanism. With the firing mechanism mounted on the cannon, two primers shall be fired by the firing mechanism of each cannon: one by a rearward pull of the lever, and one by a rotation of the case assembly. After firing each primer, the fired primer case shall be extracted by the extractor. Evidence of binding, interference, erratic movement, malfunction and misfires caused by the firing mechanism shall be cause for rejection.

4.5.4.3 Breech mechanism. The breech of each cannon shall be operated manually to determine the smoothness of operation by opening and closing the breech for two cycles. The proper functioning of the breech mechanism shall be checked by working the operating lever and observing the operation of the breechblock, breechblock-roller, breech ring, cam, counterbalance, and latching and unlatching of the operating lever. Any evidence of binding or erratic operation shall be cause for rejection. The bearing of the obturator parts shall be inspected in accordance with MIL-C-13931.

4.5.5 Performance tests. Cannons and repair tubes shall be subject to acceptance testing by a Government testing activity to assure compliance with 3.6. Acceptance testing shall be in accordance with procedures specified by the Quality Assurance element of the procuring activity.

4.5.5.1 Material soundness. After final acceptance testing, the breechblock, bushing, breech ring, carrier, and the inner and outer surfaces of the tube shall be magnetic particle tested in accordance with the applicable drawings and specifications. The inner surfaces of the tube shall be examined by means of a borescope. Material not conforming to the requirements of 3.6.2 or to applicable drawings and specifications shall be rejected.

4.5.5.2 Bore enlargement. After firing no more than eight rounds, the bore diameter across the lands of each cannon shall be measured with an air gage, a star gage, or other approved equivalent gage. Enlargement exceeding the limits specified in Table I shall be cause for rejection.

4.5.5.3 Records. Cannons or repair tubes that pass the firing tests shall be furnished with applicable data on DA Form 2408-4 by the proving ground. Cannons or repair tubes accepted on a proof sampling basis shall be furnished with applicable data on DA Form 2408-4 by the contractor and approved by the Government representative. The responsibility of the contractor for cannons or repair tubes which fail the firing tests shall be as specified in the contract (see 6.1.1).

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4.5.6 Inspection of preparation for delivery.

4.5.6.1 Packaging inspection. Unless otherwise specified, inspection to determine compliance with cleaning, preservation, packaging, packing and marking requirements of the applicable packaging data sheets for the level designated in the contract shall be as specified in MIL-P-14232.

5. PREPARATION FOR DELIVERY.

5.1 First article package. On each contract, a first article package shall be provided in accordance with 3.1. The first article package for the cannon shall be unit packaged for the level of protection specified in the contract and packed level C. Unit packaging and packing shall be in accordance with MIL-P-14232 and Packaging Data Sheet P7238947. Marking shall be in accordance with MIL-STD-129.

5.2 Unit packaging, packing and marking, of the cannon. The cannon shall be unit packaged and packed in accordance with the requirements of MIL-P-14232 and Packaging Data Sheet P7238947 for the level of protection specified in the contract (see 6.1). Marking shall be in accordance with MIL-STD-129.

5.3 Unit packaging, packing and marking of repair parts. Repair parts shall be unit packaged and packed in accordance with the requirements of MIL-P-14232 and the applicable Packaging Data Sheets for the level of protection specified in the contract (see 6.1). Marking shall be in accordance with MIL-STD-129.

6. NOTES.

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number and date of this specification

b. Requirements for submission of and shipping instructions for first article (see 3.1 and 4.4) and first article package. (see 3.1 and 5.1).

(1) The contract should designate the examinations and tests to be performed by the contractor and the examinations and tests to be performed by the Government.

(2) The contract should specify the sample size (i.e., the number of units) of the first article.

(3) The contract should designate the inspection and test data that is required to be furnished the Government when the contractor is required to perform either or both examinations and tests.

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6.1.1 cont'd. (4) The contract should designate at whose expense a retest may be performed.

(5) The contract should specify that the acceptable first article shall be delivered in accordance with the terms of the contract, or

(6) Any additional or extended examinations and tests beyond the scope of this specification should be specified and identified as a separate item of the contract.

c. The serial numbers for the cannons and repair tube assemblies.

d. List of inspection equipment, responsibility for acquisition, maintenance, and disposition thereof, if other than as specified (see 4.5.2).

e. Availability of inspection equipment from the Government.

f. Extent of supplier's responsibility for Government-furnished and for supplier-required final inspection equipment.

g. Applicable acceptance test procedures and frequency of testing (see 4.5.4).

h. The contract should specify under what conditions simulated firing may be performed, the test equipment available, and the procedures to be followed.

i. When applicable, the contract should specify the sampling plan for the actual firing of repair tubes.

j. The levals of preservation, packaging, and packing required (see 5.2).

k. Packaging instructions for repair parts (see 5.3).

6.1.2 Contract data requirements. Monthly reports of the results of final examination and performance testing shall be as specified for delivery on DD Form 1423 in the contract.

6.2 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate, on the Management Control Systems Summary List DD Form 1660.

6.3 Unless otherwise specified, the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control Systems Summary List, DD Form 1660.

Custodian:
Army - WV

Preparing activity
Army - WV

Project No. 1025-A018

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-11605B(WV) Cannon, 155mm Gun M46

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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